A REMARK ON GENERALIZED HAMMING WEIGHTS OF COMPLETE-INTERSECTION CODES: THE PLANAR AND SPLIT CASES

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Abstract. Camps Moreno, Salizzoni and San-José conjectured (arXiv:2608.19978, Conjecture 3.1) that among evaluation codes $C_d(X)$ on reduced complete intersections $X \subset \mathbb{P}^n(\mathbb{F}_q)$ of a fixed type, the projective Cartesian code has the smallest r-th generalized Hamming weight, for every d and every r. For $r = 1$ this is known to be equivalent to the Cayley–Bacharach conjecture CB12 of Eisenbud–Green–Harris. We record that the same residual argument works for every r: the Eisenbud–Green–Harris (EGH) conjecture for the type of X implies the inequality for all r and d. Since EGH is a theorem in two variables and when all but the first two generators split into linear forms, the conjecture holds, for all r and d and over every $\mathbb{F}_q$, for every reduced complete intersection in $\mathbb{P}^2$ and for every complete intersection whose generators of the $n-2$ largest degrees are products of linear forms. The argument is an assembly of known ingredients; we claim nothing beyond these two cases.

1. Statement

Let q be a prime power, $S = \mathbb{F}_q[x_0, \dots, x_n]$, and let $X = \{P_1, \dots, P_N\} \subset \mathbb{P}^n(\mathbb{F}_q)$ be a set of $\mathbb{F}_q$-rational points whose vanishing ideal is a complete intersection, $I_X = (f_1, \dots, f_n)$ with f_i homogeneous of degree a_i and $2 \le a_1 \le \cdots \le a_n$; then $N = \prod a_i$. Put

$$s := \sum_{i=1}^{n} (a_i - 1), \qquad B(a) := \prod_{i=1}^{n} \{0, 1, \dots, a_i - 1\} \subset \mathbb{Z}^n_{\ge 0}.$$

The code $C_d(X) \subseteq \mathbb{F}_q^N$ is the image of $S_d \to \mathbb{F}_q^N$, $f \mapsto (f(v_1), \dots, f(v_N))$ for fixed representatives v_j of P_j; its r-th generalized Hamming weight is $d_r(C) = \min\{|\operatorname{supp} D| : D \le C,\ \dim D = r\}$ [18].

Definition 1. A *down-set* of $B(a)$ is a subset $\mathcal{D} \subseteq B(a)$ with $\beta \in \mathcal{D}$, $\beta' \le \beta$ (componentwise) $\Rightarrow \beta' \in \mathcal{D}$. For $m \in \mathbb{Z}$ and $r \ge 1$ put

$$\mu_r(a, m) := \min\big\{|\mathcal{D}| : \mathcal{D} \subseteq B(a) \text{ a down-set},\ \#\{\beta \in \mathcal{D} : |\beta| > m\} \ge r\big\},$$

where $|\beta| = \sum \beta_i$ (and $\mu_r = \infty$ if no such $\mathcal{D}$ exists).

For $A_1, \dots, A_n \subseteq \mathbb{F}_q$ with $|A_i| = a_i$ (so $q \ge a_n$), the projective Cartesian set is $X_{\mathrm{Car}} = \{[1 : \alpha_1 : \cdots : \alpha_n] : \alpha_i \in A_i\}$, a reduced complete intersection of type a with $f_i = \prod_{\alpha \in A_i} (x_i - \alpha x_0)$.

We say that *EGH holds for X* if the following holds over $K = \overline{\mathbb{F}}_q$: for every linear form $\ell \in K[x]_1$ that does not vanish at any point of X, every homogeneous ideal J of $R := K[x]/(\ell) \cong K[y_1, \dots, y_n]$ containing the images $\bar{f}_1, \dots, \bar{f}_n$ has the Hilbert function of a homogeneous ideal containing $y_1^{a_1}, \dots, y_n^{a_n}$. This is implied by the EGH conjecture [10, 11] for n variables and degrees a (note that $\bar{f}_1, \dots, \bar{f}_n$ is a regular sequence in R, see the proof of Lemma 6).

Theorem 2. *If EGH holds for X, then for every $d \ge 1$ and every $1 \le r \le \dim C_d(X)$,*

$$d_r\big(C_d(X)\big) \ \ge\ \mu_r(a,\, s - 1 - d) \ =\ d_r\big(C_d(X_{\mathrm{Car}})\big).$$

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(The Cartesian code exists when $q \geq a_n$; the bound $d_r(C_d(X)) \geq \mu_r(a, s-1-d)$ holds regardless.)

Corollary 3. *Conjecture 3.1 of* [3] *holds, for every $q \geq a_n$, every $d \geq 1$ and every r, in the following cases (and $d_r(C_d(X)) \geq \mu_r(a, s-1-d)$ holds for every q):*

(G1) *$n = 2$: X is any reduced complete intersection of $\mathbb{F}_q$-points in $\mathbb{P}^2$;*
(G3) *$n \geq 3$ and $f_3, \ldots, f_n$ split into linear factors over $\overline{\mathbb{F}}_q$.*

Proof. For (G1), R has two variables, and EGH in two variables holds over every field [16, 6]. For (G3), the images $\bar{f}_3, \ldots, \bar{f}_n$ are again products of linear forms, and [1, Cor. 4.4] (stated over an arbitrary field, with $a_1 \leq \cdots \leq a_n$, so the split generators are those of the $n-2$ largest degrees; ties may be reordered) gives EGH for every homogeneous ideal of R containing $\bar{f}_1, \ldots, \bar{f}_n$. (For $n \geq 4$ the induction behind [1, Cor. 4.4] applies [1, Thm. 4.2] with EGH in $n-1$ variables only to ideals containing the images of $f_1, \ldots, f_{n-1}$ modulo a linear factor of f_n; these images are again regular with the last $n-3$ of them split, so the inductive hypothesis suffices.) Apply Theorem 2. □

Before this remark the following cases were known. For $r = 1$: in $\mathbb{P}^2$ [3, 14]; in $\mathbb{P}^3$, by combining the $r = 1$ equivalence [8, Prop. 6.8] with CB12 in $\mathbb{P}^3$ [13, Cor. 4.4], [4, Cor. A]; and for $d < a_1$ in every $\mathbb{P}^n$ [3, Thm. 2.3]. For all r: $d = 1$ [3, Thm. 2.5] (equigenerated case earlier in [8, Cor. 6.6]); $d = s-1$, where $d_r = r+1 = \mu_r(a, 0)$ [12, Prop. 9]; $d \geq s$, where the code is the whole space; and (implicitly, through the footprint bound [15, Thm. 4.9] and [2]) the case where the initial ideal of I_X is $(x_1^{a_1}, \ldots, x_n^{a_n})$. See Remark 8.

2. Proof

Lemma 4 ([15, Thm. 4.5])**.** *For $Y \subseteq X$ write $h_Y(t) = \dim_{\mathbb{F}_q}(S/I_Y)_t$. Then*

$$d_r(C_d(X)) = \min\{|\Gamma| : \Gamma \subseteq X,\ h_X(d) - h_{X\setminus\Gamma}(d) \geq r\}.$$

Proof. The kernel of $S_d \to C_d(X)$ is $(I_X)_d$, and the codewords supported in Γ form the image of $(I_{X\setminus\Gamma})_d$, of dimension $h_X(d) - h_{X\setminus\Gamma}(d)$. An r-dimensional subcode supported in Γ exists iff this dimension is $\geq r$. □

Lemma 5 (Cayley–Bacharach, [9], [11, CB7])**.** *For $\Gamma \subseteq X$ and every t, $h_X(t) - h_{X\setminus\Gamma}(t) = |\Gamma| - h_\Gamma(s-1-t)$ (with $h_\Gamma(u) = 0$ for $u < 0$).*

All Hilbert functions above are unchanged by extending scalars to $K = \overline{\mathbb{F}}_q$, since they are ranks of evaluation matrices with entries in $\mathbb{F}_q$; from now on we work over K.

Lemma 6. *Assume EGH holds for X and let $\emptyset \neq \Gamma \subseteq X$. There is a down-set $\mathcal{D} \subseteq B(a)$ with $|\mathcal{D}| = |\Gamma|$ and $h_\Gamma(m) = \#\{\beta \in \mathcal{D} : |\beta| \leq m\}$ for all m.*

Proof. Choose $\ell \in K[x]_1$ vanishing at no point of X (K is infinite). Since I_Γ is radical with associated primes the ideals of the points of Γ, ℓ is a nonzerodivisor on S_K/I_Γ, so $A := S_K/(I_\Gamma + (\ell))$ has Hilbert function $\Delta h_\Gamma(t) = h_\Gamma(t) - h_\Gamma(t-1)$, and $\sum_t \Delta h_\Gamma(t) = |\Gamma|$. Moreover $V(f_1, \ldots, f_n, \ell) = X \cap V(\ell) = \emptyset$, so $(f_1, \ldots, f_n, \ell)$ is primary to the irrelevant ideal and $f_1, \ldots, f_n, \ell$ is a regular sequence in the $(n+1)$-variable ring S_K; hence $\bar{f}_1, \ldots, \bar{f}_n$ is a regular sequence in R, of degrees a, contained in the image J of I_Γ (because $I_X \subseteq I_\Gamma$). By EGH there is a homogeneous ideal $J' \supseteq (y_1^{a_1}, \ldots, y_n^{a_n})$ with $H_{R/J'} = H_{R/J} = H_A$. Its initial ideal M (for any monomial order) has the same Hilbert function and still contains the pure powers. The exponents of the standard monomials of M form a down-set $\mathcal{D} \subseteq B(a)$, and $\#\{\beta \in \mathcal{D} : |\beta| = t\} = H_A(t) = \Delta h_\Gamma(t)$. Summing over t gives both claims. □

Proof of the inequality in Theorem 2. Let Γ attain the minimum in Lemma 4. By Lemma 5 with $m := s-1-d$, $r \le h_X(d) - h_{X\setminus\Gamma}(d) = |\Gamma| - h_\Gamma(m)$. With $\mathcal{D}$ from Lemma 6, $|\Gamma| - h_\Gamma(m) = \#\{\beta \in \mathcal{D} : |\beta| > m\}$, so $\#\{\beta \in \mathcal{D} : |\beta| > m\} \ge r$ and $d_r(C_d(X)) = |\Gamma| = |\mathcal{D}| \ge \mu_r(a, m)$. □

Proposition 7. *$d_r(C_d(X_{\mathrm{Car}})) = \mu_r(a, s-1-d)$ for all $d \ge 1$ and $1 \le r \le \dim C_d(X_{\mathrm{Car}})$.*

Proof. X_{Car} satisfies the hypothesis of (G3) for $n \ge 3$ and of (G1) for $n = 2$ (and EGH is trivial for $n = 1$), so "$\ge$" follows from the first part. For "$\le$", enumerate $A_i = \{\alpha_{i,0}, \dots, \alpha_{i,a_i-1}\}$ and identify $B(a)$ with X_{Car} via $\beta \mapsto P_\beta = [1 : \alpha_{1,\beta_1} : \cdots : \alpha_{n,\beta_n}]$. Let $\mathcal{D}$ attain $\mu_r(a, m)$, $m = s-1-d$, and let $\Gamma = \{P_\beta : \beta \in \mathcal{D}\}$. For $\gamma \in B(a)$ put

$$G_\gamma := x_0^{d-s+|\gamma|} \prod_{i=1}^{n} \prod_{j=\gamma_i+1}^{a_i-1} (x_i - \alpha_{i,j} x_0), \qquad \deg G_\gamma = d \ \text{ when } |\gamma| \ge s-d, \text{ i.e. } |\gamma| > m.$$

$G_\gamma(P_\beta) \ne 0$ iff $\beta \le \gamma$. For $\gamma \in \mathcal{D}$ with $|\gamma| > m$, G_γ therefore vanishes outside $\{P_\beta : \beta \le \gamma\} \subseteq \Gamma$ (down-set), and these codewords are linearly independent (order them by a linear extension of $\le$; the matrix $(G_\gamma(P_\beta))$ is triangular with nonzero diagonal). They span a subcode of dimension $\ge r$ supported in Γ, so $d_r(C_d(X_{\mathrm{Car}})) \le |\mathcal{D}|$. □

This recovers, in the present normalization, the generalized Hamming weights of affine Cartesian codes computed by Beelen and Datta [2] (the code $C_d(X_{\mathrm{Car}})$ is the affine Cartesian code of polynomials of degree $\le d$).

3. Remarks

Remark 8 (What was known, and how small the step is). (i) For $r = 1$, the equivalence of the minimum-distance statement with CB12 of [11] is [8, Prop. 6.8] (also noted in [3, §3]); EGH implies CB12 by the Artinian-reduction argument of [11]. Theorem 2 is that same argument run with "$\ge r$" in place of "≥ 1". The only point needing care is that the EGH output must be read as a down-set *inside the box* $B(a)$ whose profile above degree m is counted, and that the Cartesian code attains the resulting bound (Proposition 7). (ii) Hilbert functions of subsets of complete intersections via EGH were studied by Cooper [7]; the $r = 1$ statement (CB12) for complete intersections in $\mathbb{P}^3$ is [13, Cor. 4.4] and [4, Cor. A], the latter with a partial result in $\mathbb{P}^n$ [4, Thm. B]. (iii) If the initial ideal of I_X is $(x_1^{a_1}, \dots, x_n^{a_n})$, the footprint bound [15, Thm. 4.9] already gives Theorem 2's inequality for all r. (iv) Any further known case of EGH transfers in the same way, e.g. the degree conditions of [6] and of [5, Thm. A] ($a_i \ge \sum_{j<i}(a_j - 1)$ for $i \ge 3$, which would cover all reduced complete intersections of types $(2,2,2)$, $(2,2,3)$, $(2,3,3)$, $(2,3,4)$ in $\mathbb{P}^3$), or Cooper's cases in three variables. We have not audited the field hypotheses of these and claim only (G1) and (G3).

Remark 9 (Relation to OpenAI family 200). The manuscript *Commuting Division-Coefficient Forms and the Artinian Eisenbud–Green–Harris Conjecture* in github.com/openai/math (family 200) claims EGH over fields of characteristic 0. Codes live in characteristic p, so that claim gives nothing for Conjecture 3.1 over $\mathbb{F}_q$, and Corollary 3 does not use it. What Theorem 2 does show is that a characteristic-free proof of EGH would settle Conjecture 3.1 for all r and d; in characteristic 0, if family 200 is correct, the same argument gives the corresponding inequality for the generalized minimum distance function [15] of every reduced complete intersection of K-points, K of characteristic 0. We have not verified family 200.

Remark 10 (Strictness). The inequality can be strict for non-Cartesian X: in the computations below, type $(3,4)$ in $\mathbb{P}^2(\mathbb{F}_{13})$ produced pairs (d, r) with $d_r(C_d(X)) > \mu_r$.

4. Computations

Two independent programs (a Sage brute force over supports, and a pure-Python program using the column matroid and exhaustive down-set enumeration, with complete intersections certified by a Hilbert-function test and an exact count of rational points) compared $d_r(C_d(X))$ with $\mu_r(a, s-1-d)$ for random reduced complete intersections of types $(2,2)$–$(4,4)$ and $(2,5)$, $(3,5)$ in $\mathbb{P}^2$, and types $(2,2,2)$ (not necessarily split), and split types $(2,2,2)$, $(2,2,3)$, $(2,3,3)$ in $\mathbb{P}^3$, over prime fields $\mathbb{F}_7$–$\mathbb{F}_{17}$, for all $1 \le d \le s$ and all r: $1050 + 1424$ pairs (d, r). A third program, written by the adversarial reviewer, added exhaustive searches over $\mathbb{F}_2$, $\mathbb{F}_3$, $\mathbb{F}_4$ and random instances over $\mathbb{F}_4$, $\mathbb{F}_5$, $\mathbb{F}_8$, $\mathbb{F}_9$ and larger fields, including reducible curves and unions of lines (3011 instances, 38 189 pairs (d, r)). No violation was found, Cartesian codes attained μ_r in every case, and the deliberately mis-shifted bound $\mu_r(a, s-d)$ was violated, confirming that the test detects the shift. Lemma 5 and the down-set conclusion of Lemma 6 were also tested directly on random subsets. Scripts and outputs accompany this note.

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